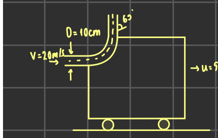
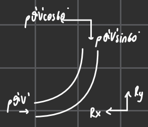
Impulse-Momentum Questions

Q1: Find the force exerted by the water jet on the object





$V' = (V-u) = 20-5 = 15 m/s$ (relative velocity)

$Q' = V'.A = 15 \times \frac{\pi \times 0.1^2}{4} = 0.1178 m^3/s$ (relative discharge)

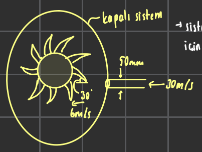
$\overset{+}{\rightarrow} \Sigma F_x = 0 = \rho Q' V' - \rho Q' V'.\cos 60 = R_x$

$1 \times (0.1178) \times 15 - 1 \times (0.1178) \times 15 \times \cos 60° = 0.8835 kN = 883.5 N = R_x$

$\overset{+}{\uparrow} \Sigma F_y = 0 = \rho Q' V' \sin 60° = R_y$ $R_y = 1.53 kN = 1530 N$ $R = \sqrt{0.8835^2 + 1.53^2} = 1.767 kN$

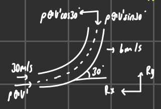
Q2: The speed of the water jet observed in the figure is 20m/s and it's diameter is 50mm. It effects the blade of a wheel turbine. Find the force acting on this turbine wheel.

~ kanatçığa etkiyen kuvvet



→ sistemde debi değişmeyeceği için rolatif debi yoktur

└ türbinler için geçerli



$Q = V.A = 30 \times \frac{\pi \times 0.05^2}{4} = 0.0589 m^3/s$

$V' = (V-u) = 30-6 = 24 m/s$

$\overset{+}{\rightarrow} \Sigma F_x = 0 = \rho Q V' - \rho Q V' \cos 30° = R_x \Rightarrow R_x = 0.189 kN = 189 N$

$\overset{+}{\uparrow} \Sigma F_y = 0 = \rho Q V' \sin 30° = R_y \Rightarrow R_y = 0.7068 kN = 706.8 N$

$R = \sqrt{0.189^2 + 0.7068^2} = 0.732 kN$

Dimensional Analysis

Prob.1- Obtain an expression for pressure gradient in a circular pipe flow.

$$f_1(\underbrace{\Delta p, L, \rho, V, D, \mu, k}_{variables}) = 0$$

Δp: N/m²; L: m; ρ: kg/m³; V: m/s; D: m; μ: dynamic viscosity, Ns/m²; k: effective roughness, m

$\tau = \mu \frac{du}{dy}$ (N/m^2), du (m/s), dy (m)

N: kg.m/s²

* önce hepsini 3 boyutta aynı birimlerle yaz (kütle, uzunluk, zaman; kg, m, s)

n: number of variables
m: number of main dimensions
} number of dimensionless parameters (pi term): n-m = 7-3 = 4 pi terms

$$\frac{\Delta p}{\rho v^2} = \frac{L}{D}.f(\overbrace{\pi_3, \pi_4}^{\lambda \to Re, k/D}) = \frac{\Delta p}{\rho g} = \frac{LV^2}{gD}$$

└ sınavda buraya kadar yazmamız yeterli

$\Delta h_L = \lambda \frac{L}{D} \frac{V^2}{2g}$

repeating variables should be choosen

Fluid (ρ, μ)
Flow (V)
Flow geometry (D, L)

$f_1(\Delta p, L, \rho, V, D, \mu, k)$ — repeating variables: ρ, V, D

$f_2(\pi_1, \pi_2, \pi_3, \pi_4)$

ρ = kg/m³ = M.L⁻³
D = m = L
V = m/s = L.T⁻¹

dimensionless:

$\pi_1 = \rho^{x_1}.D^{y_1}.V^{z_1}.\Delta p = M^0 L^0 T^0$ (mass, lenght, time) $= \underbrace{(M.L^{-3})^{x_1}.(L)^{y_1}.(L.T^{-1})^{z_1}.(M.L^{-1}T^{-2})}$

$\pi_2 = \rho^{x_2}.D^{y_2}.V^{z_2}.L$

$\pi_3 = \rho^{x_3}.D^{y_3}.V^{z_3}.k$

$\pi_4 = \rho^{x_4}.D^{y_4}.V^{z_4}.\mu$

for M: $x_1 + 1 = 0 \Rightarrow x_1 = -1$

for L: $-3x_1 + y_1 + z_1 - 1 = 0 \Rightarrow y_1 + z_1 = -2$

for T: $-z_1 - 2 = 0 \Rightarrow z_1 = -2$, $y_1 = 0$

→ apply these steps for the remaining pi terms

$\pi_1 = \rho^{-1}.V^{-2}.\Delta p = \frac{\Delta p}{\rho.V^2}$

$\pi_2 = D^{-1}.L = L/D$

$\pi_3 = k/D$

$\pi_4 = \rho^{-1}.D^{-1}.V^{-1}.\mu = \mu/\rho DV = \nu/DV = 1/Re$

└ reynold's number

Q2: Determine the resistance force expression of the D diameter.

$f_1(F, D, V, \rho, \mu)$ — F: aranan; D: L; V: L.T⁻¹; ρ: M.L⁻³

n=5
m=3

#pi terms = n-m = 2

$\pi_1 = D^{x_1}.\rho^{y_1}.V^{z_1}.F \to M^0 L^0 T^0 = (L)^{x_1}.(M.L^{-3})^{y_1}.(L.T^{-1})^{z_1}.(M.L.T^{-2})$

for M: $y_1 + 1 = 0$, $y_1 = -1$
for T: $-z_1 - 2 = 0$, $z_1 = -2$
for L: $x_1 - 3y_1 + z_1 + 1 = 0$, $x_1 = -2$
} $\pi_1 = D^{-2}.\rho^{-1}.V^{-2}.F = F/\rho D^2 V^2$

$\pi_2 = D^{x_2}.\rho^{y_2}.V^{z_2}.\mu \to M^0 L^0 T^0 = (L)^{x_2}.(M.L^{-3})^{y_2}.(L.T^{-1})^{z_2}.(M.L^{-1}.T^{-1})$